

Mike P. Wittie

Distributed Systems Architect & Technical Lead

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I work at the intersection of systems research and engineering, designing and deploying distributed systems from first principles through production by reducing complex problems into simple, composable, measurable primitives. As a former founder, CEO, Chief Scientist, tenured professor, and currently Staff Engineer, I focus on identifying the key constraint in a complex system, targeting it directly, and measuring the resulting performance in deployment. I operate as a deep technical individual contributor or hands-on technical lead, building systems using expertise in confidential computing, blockchain systems, and latency-sensitive environments to distill trade-offs into actionable decisions that enable deployment and adoption.

Employment

- Staff Engineer, Arkenstone Defense 2026 – present
 - Led the development of a PEO employee portal integrating ProSoftware (HRIS), LMNTL (time-keeping), Carta (equity), and Litmos (LMS).
 - Accelerated team velocity by designing and driving the adoption of AI-driven development lifecycle (AIDLC) workflows spanning product definition, implementation planning, delivery, and deployment, integrated with Jira Product Discovery, Jira, and Confluence and backed by reusable agent skills and architecture decision records (ADRs).
- Chief Executive Officer, Blocky, Inc. 2025 – 2026
 - Designed and executed product strategy for a verifiable serverless compute platform (WASM on confidential computing), enabling secure off-chain execution with reduced trusted computing base and startup latency vs. Chainlink Functions.
 - Matured product from PoC to production in 6 months, leading productization across CLI, developer portal, documentation, and billing systems.
 - Designed and ran data-driven GTM experiments (LinkedIn, X), improving outbound performance by 341% and establishing initial positioning.
 - Acquired paid design partners and enabled production time-weighted average price (TWAP) oracle deployments (Casper's [Styks](#)) validating real-world platform usage.
 - Raised \$500K+ in convertible financing to extend runway and deploy a paid beta in production.
- Chief Scientist, Blocky, Inc. 2018 – 2025
 - Led the development of successful NSF SBIR Phase I & II grants to secure \$1.25M in non-dilutive R&D funding.
 - Adapted authentication services (Auth0, Cognito) to create existence proofs and combined them with Trusted Execution Environment (TEE) attestations (AWS Nitro Enclaves, GCP Confidential Spaces) and a novel datastructure to guarantee agreement without relying on costly consensus mechanisms.
 - Architected and contributed to the implementation of [ZipperChain](#) to reduce transaction ordering cost by four orders of magnitude.
 - [Patented](#) ZipperChain in U.S., E.U., and other international jurisdictions.
 - Contributed to the implementation of ZipperChain algorithms and infrastructure, achieving 18K TPS to make it one of the fastest blockchain systems at the time.
 - Developed core technical architecture and prototypes underpinning \$2M seed raise at \$20M post-valuation.

- Assistant and Associate Professor, Montana State University 2011 – 2022
 - Led the Networks and Systems Lab, mentoring students and collaborators across multiple projects from ideation through implementation, publication, and grant development.
 - Designed and evaluated distributed systems to solve problems in content delivery, delay-sensitive communication, blockchain oracles, edge computing, and human-computer interaction (HCI) with a focus on rigorous measurement and data-driven optimization.
 - Translated research into production infrastructure deployed with industry partners including Akamai and Beartooth.
 - Published over 50 academic articles cited over 1,200 times.
 - Secured \$6.7M in competitive NSF and federal research funding, with over \$1.6M directly supporting my lab’s personnel and research initiatives.
- Consultant, Beartooth Inc., Bozeman, MT 2017 – 2021
 - Designed and implemented a real-time mesh MAC protocol for LoRa radios.
 - Advised on supporting real-time voice transmission protocol implementation.
- Earlier Industry Experience 2004 – 2008
 - Held research and engineering roles at Bell Labs, Lockheed Martin, and Rockwell Collins, working on network protocol design and data-link interoperability.

Skills

- Distributed Systems & Infrastructure:
Distributed systems architecture, consensus and replication, storage systems, networking (incl. wireless), cloud and edge platforms, fault-tolerant and latency-sensitive systems
- Production Systems & Developer Platforms:
Design and productization of distributed systems, APIs/CLIs, and developer platforms, Infrastructure-as-Code (Terraform, Ansible, Pulumi), Kubernetes orchestration, GitOps-driven CI/CD pipelines (Atlantis, Nix)
- Trust, Security & Verification:
Confidential computing and secure enclaves (AWS Nitro Enclaves, Google Confidential Spaces), remote attestation, cryptographic proofs and authenticated data structures, applied cryptography, blockchain systems
- Performance & Systems Engineering:
Performance measurement, system optimization, experimental design and evaluation
- Technical Stack:
Go, Python, C/C++; Linux; AWS, GCP
- Languages:
English (native), Polish (native), Spanish (conversational), French (basic)

Education

- Ph.D., Computer Science, University of California at Santa Barbara 2011
- M.S., Computer Science, University of Pennsylvania 2004
- B.A., Cognitive Science, University of Pennsylvania 2003